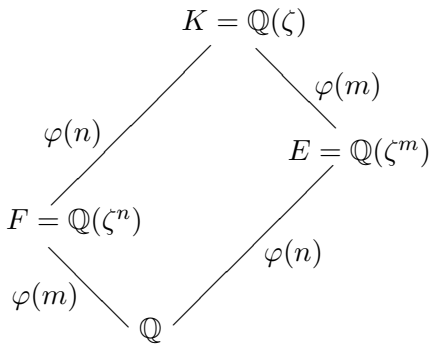
The splitting field of $(x^m-1)(x^n-1)$ contains ζ^n and ζ^m and thus contains $\zeta^m\zeta^n$. Now $\zeta^m\zeta^n=\zeta^{m+n}$ has order the least positive integer r such that mn divides $r(m+n)$. No prime dividing m divides $m+n$ because m and n are relatively prime. Similarly, no prime dividing n divides $m+n$. Consequently, mn must divide r, so $\zeta^m\zeta^n$ is a primitive mnth root of unity. Thus the splitting field of $(x^m-1)(x^n-1)$ contains a primitive mnth root of unity, and thus contains the splitting field of $x^{mn}-1$. This completes the demonstration that the splitting fields of $x^{mn}-1$ and of $(x^n-1)(x^m-1)$ are the same.

15. Let ζ be a primitive mnth root of unity, so that $K=\mathbb{Q}(\zeta)$ is the splitting field of $x^{mn}-1$. Now form $F=\mathbb{Q}(\zeta^n)$ and $E=\mathbb{Q}(\zeta^m)$ as shown in the diagram.



Now ζ^m is a primitive nth root of unity and ζ^n is a primitive mth root of unity. Thus $[F:\mathbb{Q}]=\varphi(m)$ and $[E:\mathbb{Q}]=\varphi(n)$ as labeled on the lower part of the diagram. Formula(1) for $\varphi(n)$ in the text shows that because m and n are relatively prime, $\varphi(mn)=\varphi(m)\varphi(n)$. Because $[K:\mathbb{Q}]=\varphi(mn)$, we see that $[K:F]=\varphi(n)$ and also that $[K:E]=\varphi(m)$ as labeled on the upper part of the diagram. Thus $G(K/F)$ is a subgroup of $G(K/\mathbb{Q})$ of order $\varphi(n)$ and $G(K/E)$ is a subgroup of order $\varphi(m)$.

We check the conditions of Exercise 50 of Section 11 to show that $G(K/\mathbb{Q})\simeq G(K/F)\times G(K/E)$. Because $G(K/\mathbb{Q})$ is abelian (see Theorem 55.4), condition (b) holds. For condition (c), suppose that $\sigma\in G(K/\mathbb{Q})$ is in both $G(K/F)$ and $G(K/E)$. Then $\sigma(\zeta^m)=\zeta^m$ and $\sigma(\zeta^n)=\zeta^n$. Suppose that $\sigma(\zeta)=\zeta^r$. Then $\sigma(\zeta^m)=\zeta^{rm}=\zeta^m$ so $r\equiv 1(\text{mod } n)$. Also $\sigma(\zeta^n)=\zeta^{rn}=\zeta^n$ so $r\equiv 1(\text{mod } m)$. Because n and m are relatively prime, we see that $r\equiv 1(\text{mod } mn)$, so $r=1$ and σ is the identity automorphism. Thus $G(K/F)\cap G(K/E)$ consists of just the identity automorphism. To demonstrate condition (a) that $G(K/F)\vee G(K/E)=G(K/\mathbb{Q})$, form the $\varphi(m)\varphi(n)$ elements $\sigma\mu$ where $\sigma\in F(K/E)$ and $\mu\in G(K/F)$. We claim that these products are all distinct, so that they must comprise all of $G(K/\mathbb{Q})$. Suppose that $\sigma\mu=\sigma_1\mu_1$ for $\sigma,\sigma_1\in G(K/E)$ and $\mu,\mu_1\in G(K/F)$. Then ${\sigma_1}^{-1}\sigma=\mu_1\mu^{-1}$ is in both $G(K/E)$ and $G(K/F)$, and thus must be the identity automorphism. Therefore $\sigma=\sigma_1$ and $\mu=\mu_1$. Exercise 51 of Section 11 now shows that $G(K/\mathbb{Q})\simeq F(K/F)\times G(K/E)$.

56. Insolvability of the Quintic

1. No, The splitting field E cannot be obtained by adjoining a square root of an element of $\mathbb{Z}_2$ to $\mathbb{Z}_2$ because all elements in $\mathbb{Z}_2$ are already squares. However, K is an extension by radicals, for $x^3-1=(x-1)(x^2+x+1)$ so K is also the splitting field of x^3-1. Thus $K=\mathbb{Z}_2(\zeta)$ where ζ is a primitive cube root of unity, and $\zeta^3=1$ is in $\mathbb{Z}_2$.

2. Yes, because if α is a zero of $f(x) = ax^8 + bx^6 + cx^4 + dx^2 + e$ then α^2 is a zero of $g(x) = ax^4 + bx^3 + cx^2 + dx + e$. Because $g(x)$ is a quartic, $F(\alpha^2)$ is an extension of F by radicals, and thus $F(\alpha)$ is an extension of F by radicals.

3. T T T F T F T F F T

4. We have

$$\begin{aligned} f(x) &= ax^2 + bx + c = a(x^2 + \frac{b}{a}x) + c \\ &= a(x + \frac{b}{2 \cdot a})^2 + c - \frac{b^2}{4 \cdot a} \text{ if } 2 \cdot a \neq 0. \end{aligned}$$

Thus if $\alpha \in \overline{F}$ satisfies

$$a(\alpha + \frac{b}{2 \cdot a})^2 = \frac{b^2 - 4 \cdot ac}{4 \cdot a}$$

so that

$$\alpha + \frac{b}{2 \cdot a} = \pm\sqrt{\frac{b^2 - 4 \cdot ac}{4 \cdot a^2}} \text{ and } \alpha = \frac{-b \pm \sqrt{b^2 - 4 \cdot ac}}{2 \cdot a},$$

then α is a zero of $ax^2 + bx + c$.

5. Let α be a zero of $ax^4 + bx^2 + c$. Then α^2 is a zero of $ax^2 + bx + c$ which is solvable by radicals by Exercise 4. If $\alpha_1 = \alpha, \alpha_2, \alpha_3$, and α_4 are the zeros of $ax^4 + bx^2 + c$, then the tower of field starting with F and adjoining in sequence $\alpha_1^2, \alpha_2^2, \alpha_3^2, \alpha_4^2, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ is an extension where each successive field of the tower is either equal to the preceding field or is obtained from it by adjoining a square root of an element of the preceding field. Thus the splitting field is an extension by radicals, so the quartic is solvable.

6. We can achieve any refinement of a subnormal series by inserting, one at a time, a finite number of groups. Let $H_i < H_{i+1}$ be two adjacent terms of the series, so that H_{i+1}/H_i is abelian,and suppose that an additional subgroup K is inserted between them so that $H_i < K < H_{i+1}$. Then K/H_i is abelian because it can be regarded as a subgroup of H_{i+1}/H_i. By Theorem 34.7, H_{i+1}/K is isomorphic to $(H_{i+1}/H_i)/(K/H_i)$, which is the factor group of an abelian group, and hence is abelian. For an alternate argument, note that because H_{i+1}/H_i is abelian, H_i must contain the commutator subgroup of H_{i+1}, so K also contains this commutator subgroup and H_{i+1}/K is abelian.

7. Let $H_i < H_{i+1}$ be two adjacent groups in a subnormal series with solvable quotient groups, so that H_{i+1}/H_i is a solvable group. By definition, there exists a subnormal series

$$H_i/H_i < K_1/H_i < K_2/H_i < \cdots < K_r/H_i < H_{i+1}/H_i$$

with abelian quotient groups. We claim that the refinement of the original series at this ith level to

$$K_0 = H_i < K_1 < K_2 < \cdots < K_r < H_{i+1} = K_{r+1}$$

has abelian quotient groups at this level, for by Theorem 34.7, $K_j/K_{j+1} \simeq (K_j/H_i)/(K_{j-1}/H_i)$, which is abelian by our construction. Making such a refinement at each level of the given subnormal series, we obtain a subnormal series with abelian quotient groups.

8. a. The generalization of this to an n-cycle and a transposition in S_n is proved in the solution to Exercise 39 of Section 9.